ZCAS University

CCS1401 COMPUTING AND ICT SKILLS

FINAL EXAMINATION

TUESDAY 28TH MAY,2025

14:00HRS- 17:00 HRS

TIME ALLOWED: READING AND WRITING – THREE HOURS

UPLOADING ANSWER SCRIPT – 30 MINUTES

INSTRUCTIONS:

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer ***THREE*** questions from this section.
3. This examination paper carries a total of **100 marks**.

SECTION A: All questions are compulsory and must be answered.

Question 1

A logistics company keeps a record of delivery data in a spreadsheet. The dataset includes the number of trips, distance per trip (in kilometers), cost per kilometer, and discount offered based on total cost as highlighted in the figure below.

	A	B	C	D	E	F	G	H	I	J	K
1	Invoice Number	Subject	Number of Trips	Distance per Trip (km)	Cost per km ($)	Total Distance (km)	Initial Transport Cost ($)	Discount ($)	Final Cost ($)	VAT (16%)	Total Payable ($)
2	INV001	Pharmacy Delivery	5	10	2.5						
3	INV002	Medical Equipment	12	25	2						
4	INV003	Grocery Refill	8	15	3						
5	INV004	Office Supplies	20	40	2.8						
6	INV005	Furniture	15	30	2.2						
7	INV006	Pharmacy Delivery	10	10	2.5						
8	INV007	Medical Equipment	5	25	2						
9	INV008	Grocery Refill	15	15	3						
10	INV009	Office Supplies	4	40	2.8						
11	INV010	Furniture	3	30	2.2						
12											
13		Highest intial cost									
14		Lowest delivery trips									
15		Average distance per trip									

Complete the tasks below by writing only the appropriate Excel formulas to be pasted to similar adjacent cells.

1. Write the formula to calculate the total distance (3 marks)
2. Write the formula to calculate the initial transport. (3 marks)
3. Write the formula to calculate the discount amount, where:
   - Less than $100 → 2% discount
   - $100 to less than $500 → 5% discount
   - $500 and above → 8% discount (6 marks)
4. Write the formula to calculate the final cost after subtracting the discount from the initial cost. (3 marks)
5. Write the formula to calculate the VAT amount, assuming a fixed VAT rate of 16%. (3 marks)
6. Write the formula to calculate the total payable amount by adding VAT to the final cost. (3 marks)
7. Write a formula that calculates the total initial transport cost for all deliveries with the subject "Grocery Refill". (4 marks)
8. Write a formula to count how many deliveries had more than 10 trips. (4 marks)
9. Write a formula to find the highest initial transport cost from all deliveries. (3 marks)

10. Write a formula to find the lowest number of trips from all deliveries. (3 marks)
11. Write a formula to calculate the average distance per trip from all deliveries. (2 marks)

[Total 40 Marks]

SECTION B: Answer any THREE (3) questions.

Question 2

a) System software is usually grouped in four distinct groupings. Discuss with examples the four types of system software. **(10 marks)**

b) Bwalya would want to buy storage media for the company he works for. He is a little confused on the different storage devices on the market. Advise him on the following devices in terms of access speed, cost, robustness and storage capacity.

i. Hard drive

ii. Flash disk

iii. Magnetic tapes

iv. Optical disks **(10 marks)**

[Total 20 Marks]

Question 3

The following tables provide the structure of a simple primary school database. Use these tables to answer the questions in this section.

StudentID	Name	Age	Gender	ClassID
1	John Smith	15	Male	1
2	Jane Doe	16	Female	2
3	Bob Brown	14	Male	1
4	Mary White	15	Female	2
5	Tom Peter Green	16	Male	3

TeacherID	Name	SubjectID
1	Mr. Johnson	1
2	Ms. Smith	2
3	Mr. Lee	3

ClassID	ClassName	TeacherID
1	Math	1
2	English	2
3	Science	3

SubjectID	SubjectName
1	Math
2	English
3	Science

Required:

a) Using the four tables provided for the school draw an entity-relationship (ER) diagram that accurately represents the data model for the scenario. Be sure to include all entities, attributes, primary keys, and relationships. **(12 marks)**

b) Populate the following table highlighting the required MS Access expression for the following scenarios.

NO.	Query	Required Tables	MS Access Expression
i.	Query to retrieve students with ages between 14 and 16 taking Math class		
ii.	Query to retrieve students whose name starts with the letter 'J'		
iii.	Query to retrieve student's Name and class whose middle name is "Peter"		
iv.	Write a query to display the names of students whose second letter is 'O'.		

Note that the you are required to enter what you would enter in the MS access criteria section in Query design.

(8 marks)

[Total 20 Marks]

Question 4

Write short notes about the following features of Windows operating systems:

i. User Account Control (UAC) **(4 Marks)**

ii. Windows Defender **(4 Marks)**

iii. Task Manager **(4 Marks)**

iv. File Explorer **(4 Marks)**

v. Windows Update **(4 Marks)**

In your response, provide a brief explanation of each feature and its significance in Windows operating systems. Include the key functionalities and benefits that these features offer to users. Ensure your notes are concise and informative, demonstrating your understanding of these essential features in Windows operating systems.

[Total 20 Marks]

Question 5

For the following questions, write short answers or **copy the selected choice** from the option given. Any partly written option will be treated as a different answer from the options given. Ensure the questions are correctly labelled.

Study the screenshot below and answer question **a)** and **b) only.**

	A	B	C	D	E	F
1		Jan	Feb	Mar	Total	Average
2	Sales	120	160	210	490	245
3	Overhead	100	130	160	390	195
4	Profits	20	30	50	100	50

a) The best formula to calculate Profits for January is: (1 mark)

b) The best formula to calculate the Average for Profits is: (1 mark)

c) Cell D4 contains the formula =C1+C2. What formula will see when you paste this into cell E5? (Choose from the options below) (2 mark)

- =C1+C2
- =$C1+$C2
- =D2+D3
- =E6+E7

d) When you copy a formula that contains an absolute reference to a new location, the reference ____. (Choose from the options below) (1 mark)

- is updated automatically
- does not change

- becomes bold
- has a dotted outline in its cell

e) Company A is looking into four potential projects and will accept them if the IRR is 10% or above, as shown in cell E2.

	A	B	C	D	E
1		IRR	Accept or Reject		Decision rule
2	Project 1	12.5%	Accept		10%
3	Project 2	9.3%	Reject		
4	Project 3	8.2%	Reject		
5	Project 4	11.1%	Accept		
6					

What is the formula used in cell C2, which can be copied down to cell C3 through C5, to generate the results shown below? (Choose from the options below) (2 mark)

- =IF(B2>=E2,"Accept","Reject")
- =IF(B2>=$E2,"Accept","Reject")
- =IF(B2>=E$2,"Accept","Reject")
- =IF(B2>=E2,"Accept","Reject")

f) Study the screenshot above. After doing her budget, Lucy realizes that she cannot pay more than $3,850 per month (Monthly Payment) towards a new apartment. Which ANALYTICAL function could she utilize to calculate her new loan amount based on the screenshot above? (Choose from the options below). (1 mark)

	A	B
1	APARTMENT LOAN	
2	Montly Payment (PMT):	-$5,512.02
3	Monthly Interest Rate (RATE):	2%
4	Months to Pay Off Loan (NPER):	120
5	Amount to Loan from Bank (PV):	$250,000.00

- She could use the Solver function.
- She could use the What-If Analysis function.
- She could use the Goal Seek function.
- She could use the Data Table function.

g) Suppose the formula =D7+F8/G8-H5+2 is in cell D10. Which is the best method to work out the value required in F8 to get a value of 850 from the formula? (Choose from the options below) (2 mark)

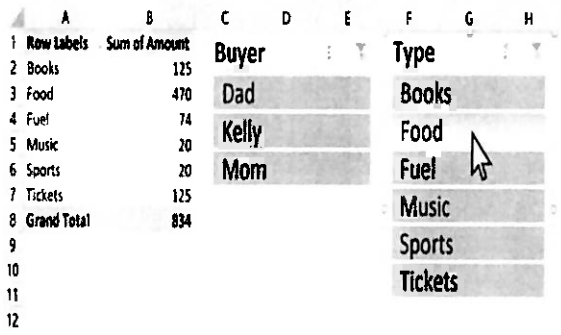
- Use Goal Seek to set D10 to 850 by changing F8
- Use the Scenario manager to place 850 in D10

- Use Goal Seek to set F8 to 850 by changing D10

h) Which of the descriptions listed below best describe what a macro in Excel does? (Choose from the options below) (1 mark)

- It is a type of formatting you can apply to enlarge the worksheet area.
- It is a file type that you save a workbook as.
- It is a series of instructions contained in one command that you can use to automate complex and/or repetitive tasks.
- It is a recording function that allows you to save worksheets, audio recordings and videos that are displayed on the screen.

i) Study the screenshot below. Name the powerful tool below used to quickly group, summarize, and rearrange larger datasets. (2 mark)



j) Which criteria return only those addresses beginning with the letter "K"? (Choose from the options below) (2 mark)

- address = "K"
- address = "K*"
- address = "K?"
- address = "K#"

k) Which criteria do not return the position "Officer" as a match? (Choose from the options below) (2 mark)

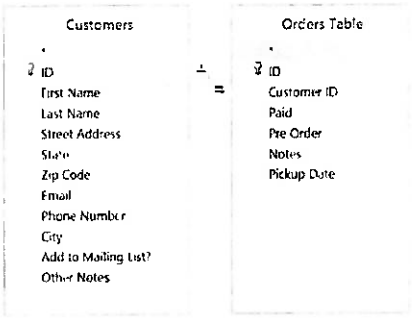
- position = "*ff*"
- position = "O??icer"
- position = " ?ff*"
- position = "O#r"

l) The ascending order of data hierarchy is? (1 mark)

- Bit - byte - field - record - file - database

- Bit - byte - record - field - file - database
- Byte - bit - field - record - file - database
- Bit - byte - field - record - database – file

m) A bakery wants to create a list of **customer names and email addresses** for customers who **pre-ordered an item and picked it up during the month of December**. The tables needed for the query are shown in this image. Aside from First Name and Last Name, what fields must be added to the query? Select all that apply. (2 mark)



- Pickup Date
- Pre Order
- Email
- City

[Total 20 Marks]

END OF EXAMINATION